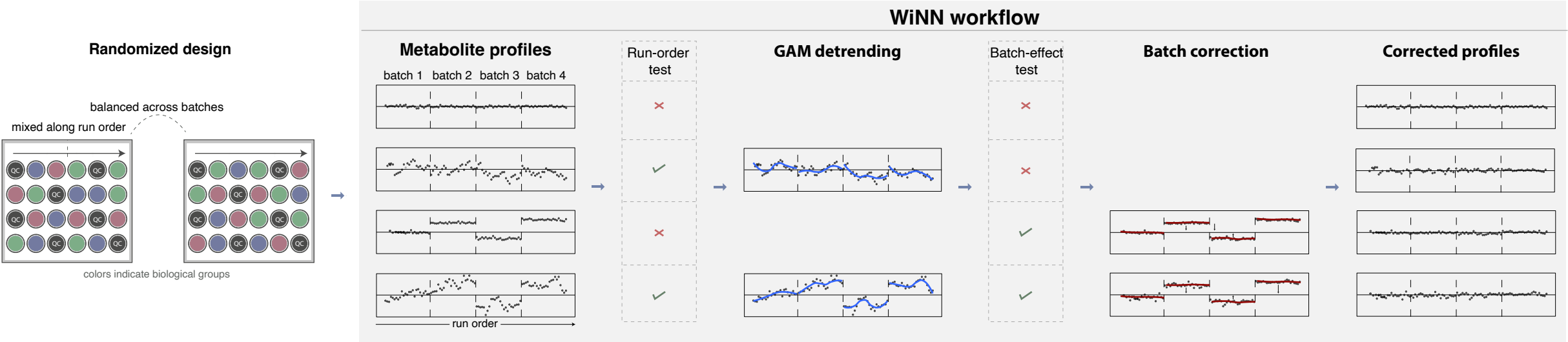
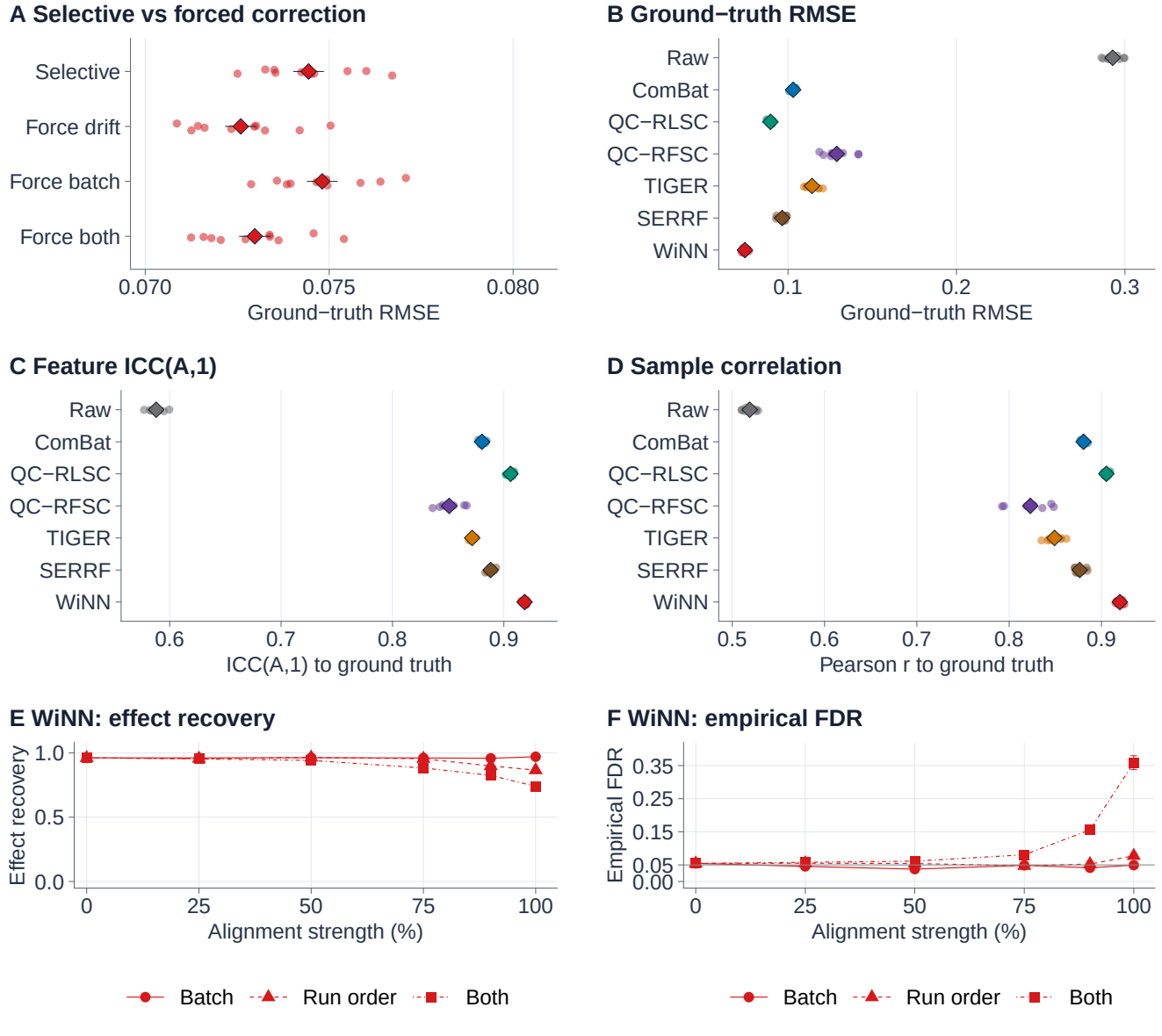
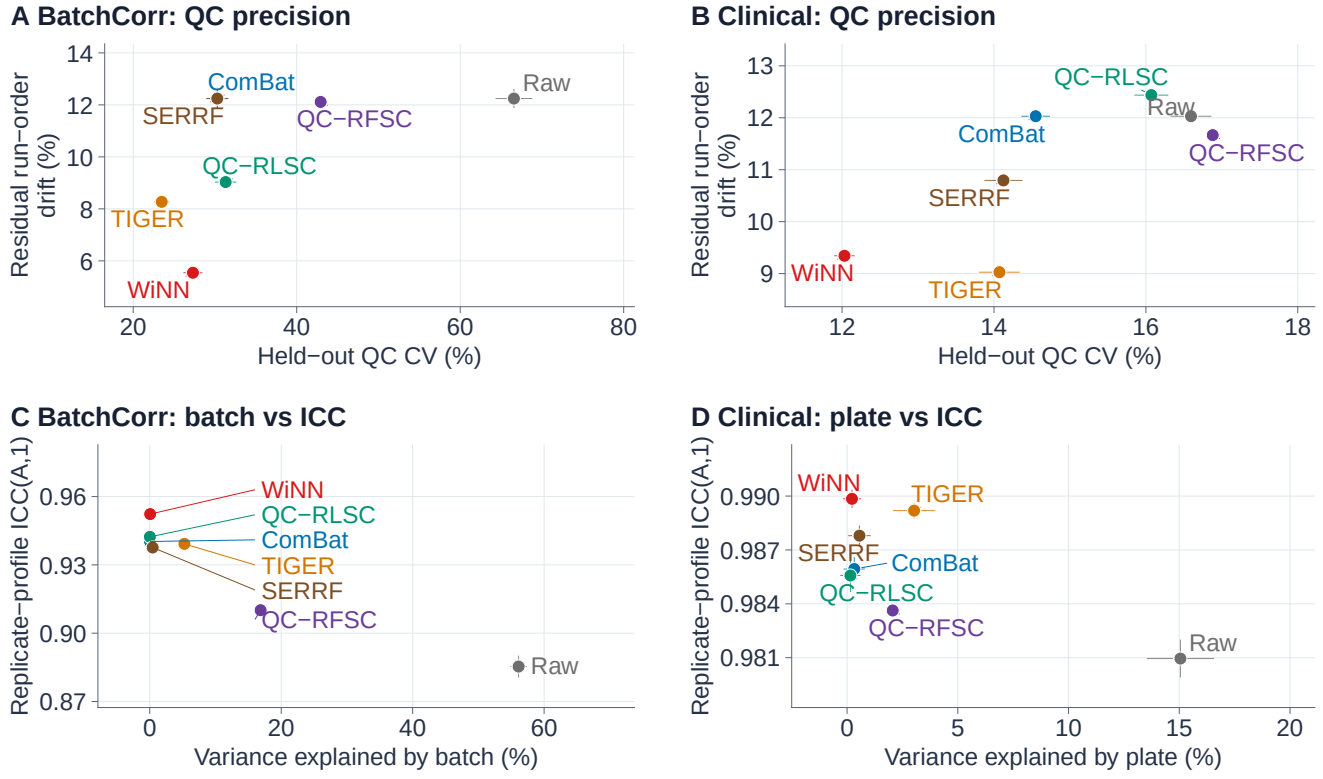
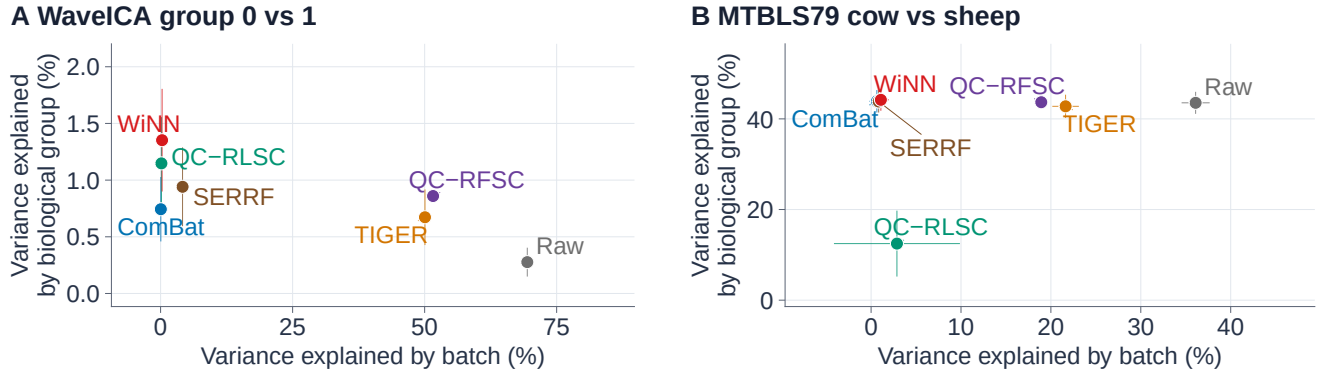
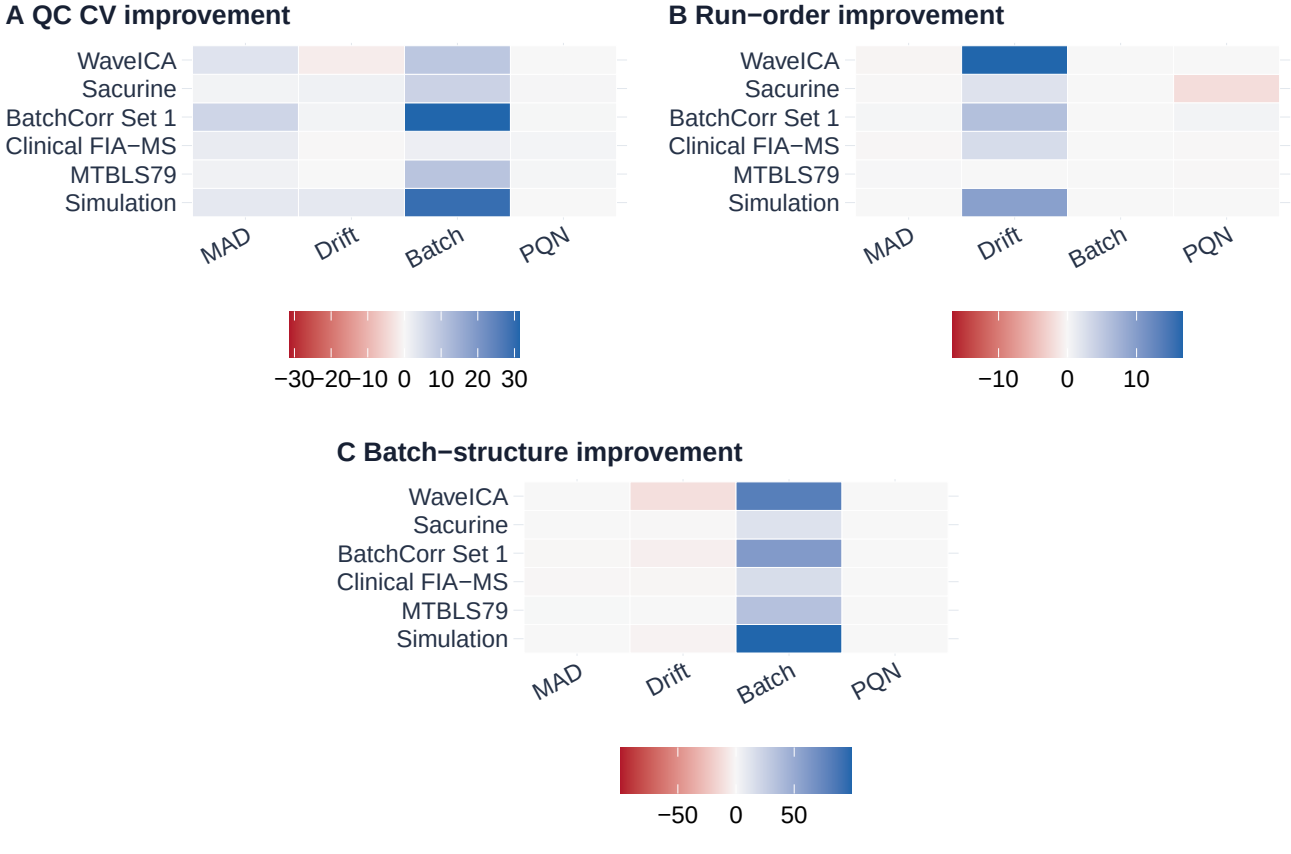
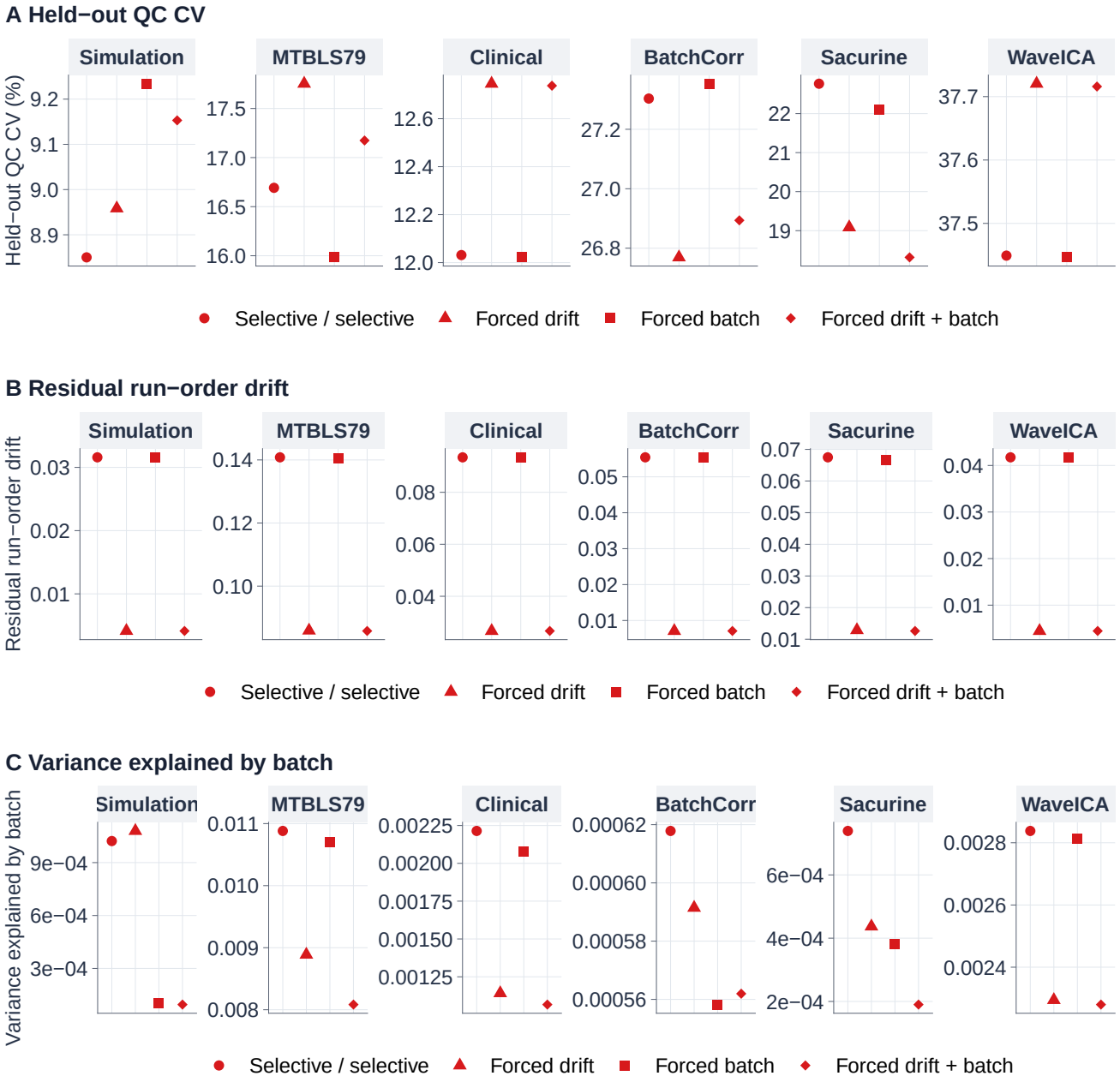

Randomized design
balanced across batches
mixed along run order
QC
colors indicate biological groups
WiNN workflow
Metabolite profiles
batch 1
batch 2
batch 3
batch 4
run order
Run-order test
GAM detrending
Batch-effect test
Batch correction
Corrected profiles

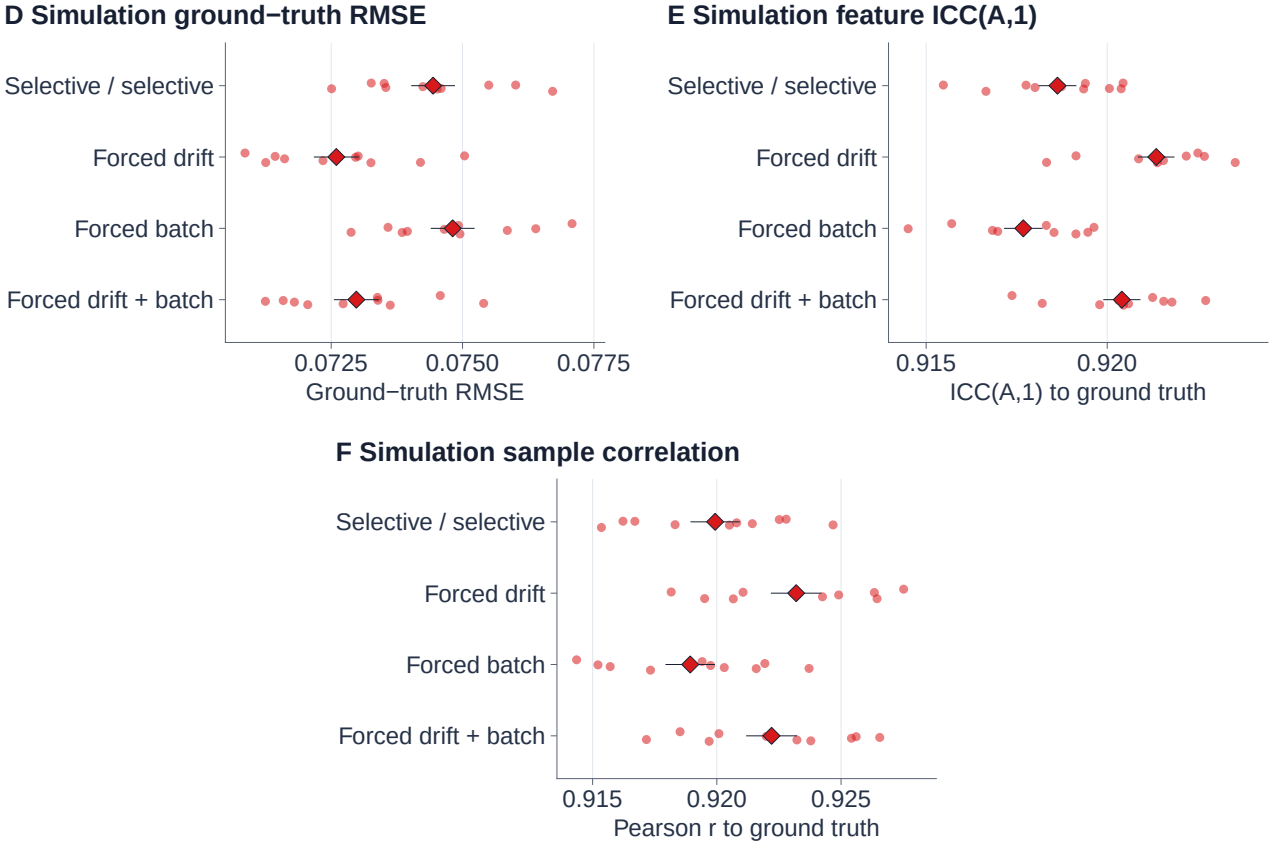

A Selective vs forced correction
Selective
Force drift
Force batch
Force both
0.070
0.075
0.080
Ground−truth RMSE
B Ground−truth RMSE
Raw
ComBat
QC−RLSC
QC−RFSC
TIGER
SERRF
WiNN
0.1
0.2
0.3
Ground−truth RMSE
C Feature ICC(A,1)
0.6
0.7
0.8
0.9
ICC(A,1) to ground truth
D Sample correlation
0.5
0.6
0.7
0.8
0.9
Pearson r to ground truth
E WiNN: effect recovery
Effect recovery
1.0
0.5
0.0
0
25
50
75
100
Alignment strength (%)
F WiNN: empirical FDR
Empirical FDR
0.35
0.25
0.15
0.05
0.00
Alignment strength (%)
Batch
Run order
Both

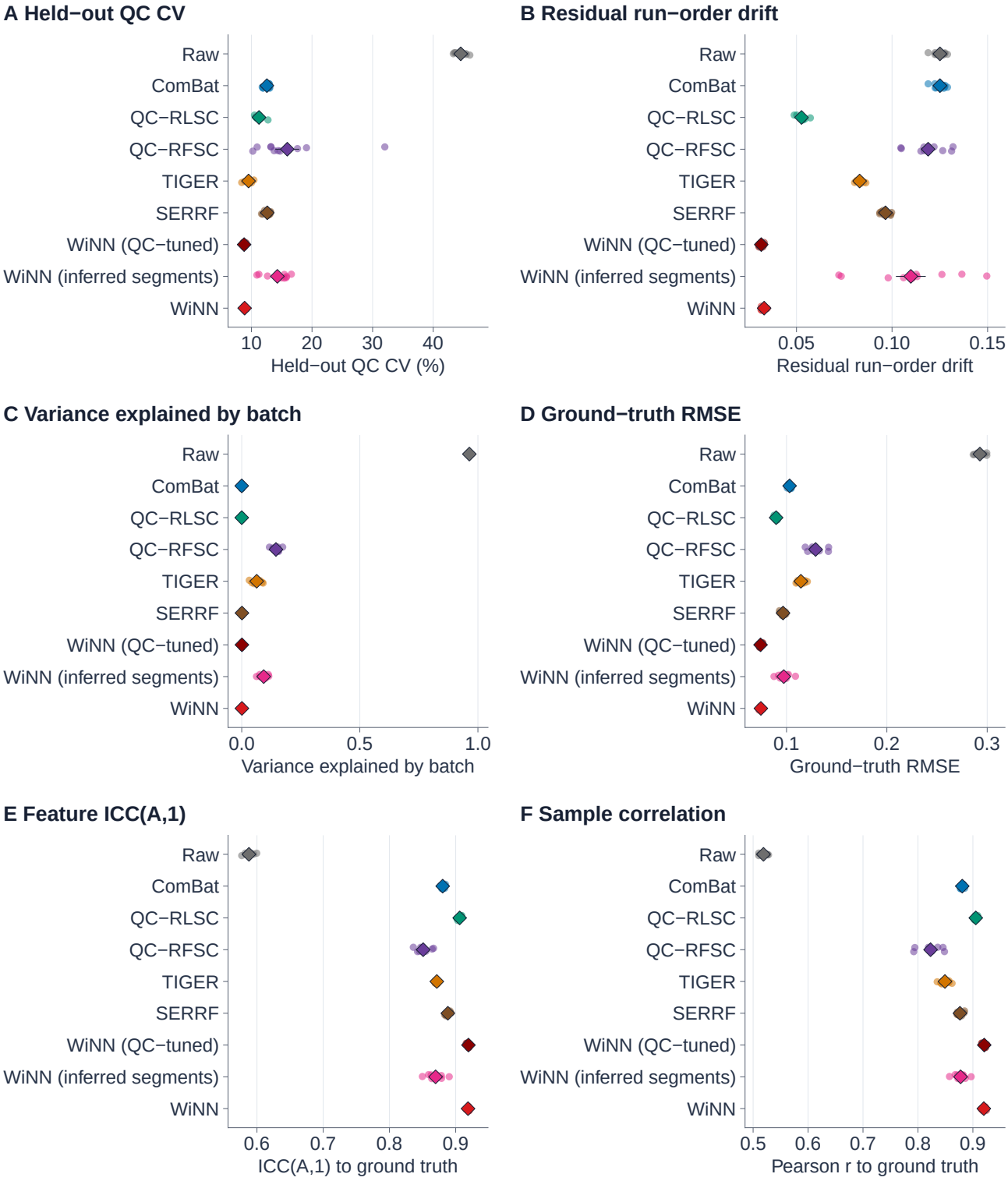

A BatchCorr: QC precision
Residual run−order drift (%)
Held−out QC CV (%)
ComBat
Raw
SERRF
QC−RFSC
QC−RLSC
TIGER
WiNN
B Clinical: QC precision
Residual run−order drift (%)
Held−out QC CV (%)
QC−RLSC
Raw
ComBat
QC−RFSC
SERRF
WiNN
TIGER
C BatchCorr: batch vs ICC
Replicate−profile ICC(A,1)
Variance explained by batch (%)
WiNN
QC−RLSC
ComBat
TIGER
SERRF
QC−RFSC
Raw
D Clinical: plate vs ICC
Replicate−profile ICC(A,1)
Variance explained by plate (%)
WiNN
TIGER
SERRF
ComBat
QC−RLSC
QC−RFSC
Raw

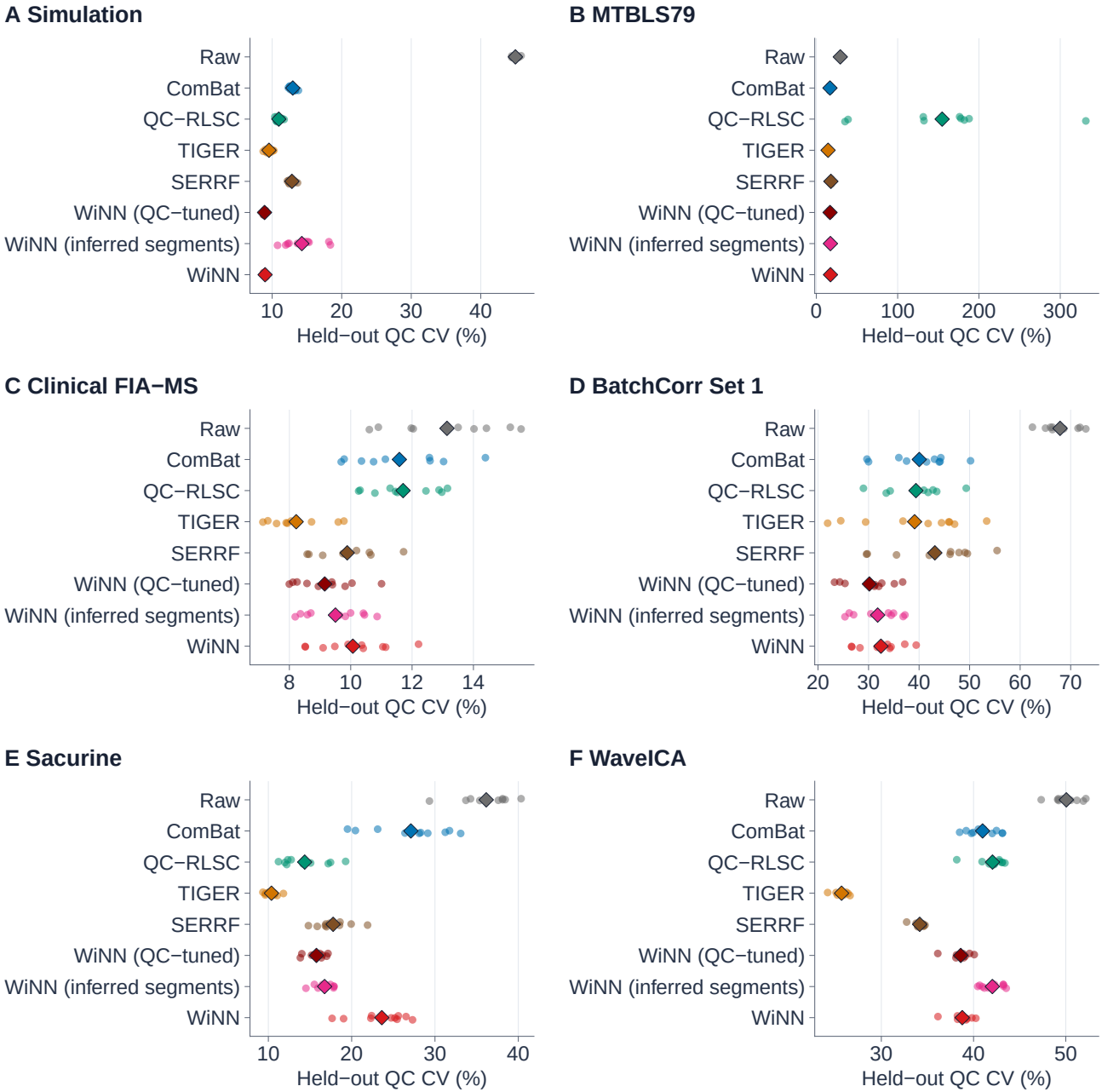

A WaveICA group 0 vs 1
Variance explained by biological group (%)
2.0
1.5
1.0
0.5
0.0
WiNN
QC−RLSC
SERRF
ComBat
QC−RFSC
TIGER
Raw
0
25
50
75
Variance explained by batch (%)
B MTBLS79 cow vs sheep
Variance explained by biological group (%)
40
20
0
WiNN
QC−RFSC
Raw
ComBat
TIGER
SERRF
QC−RLSC
0
10
20
30
40
Variance explained by batch (%)

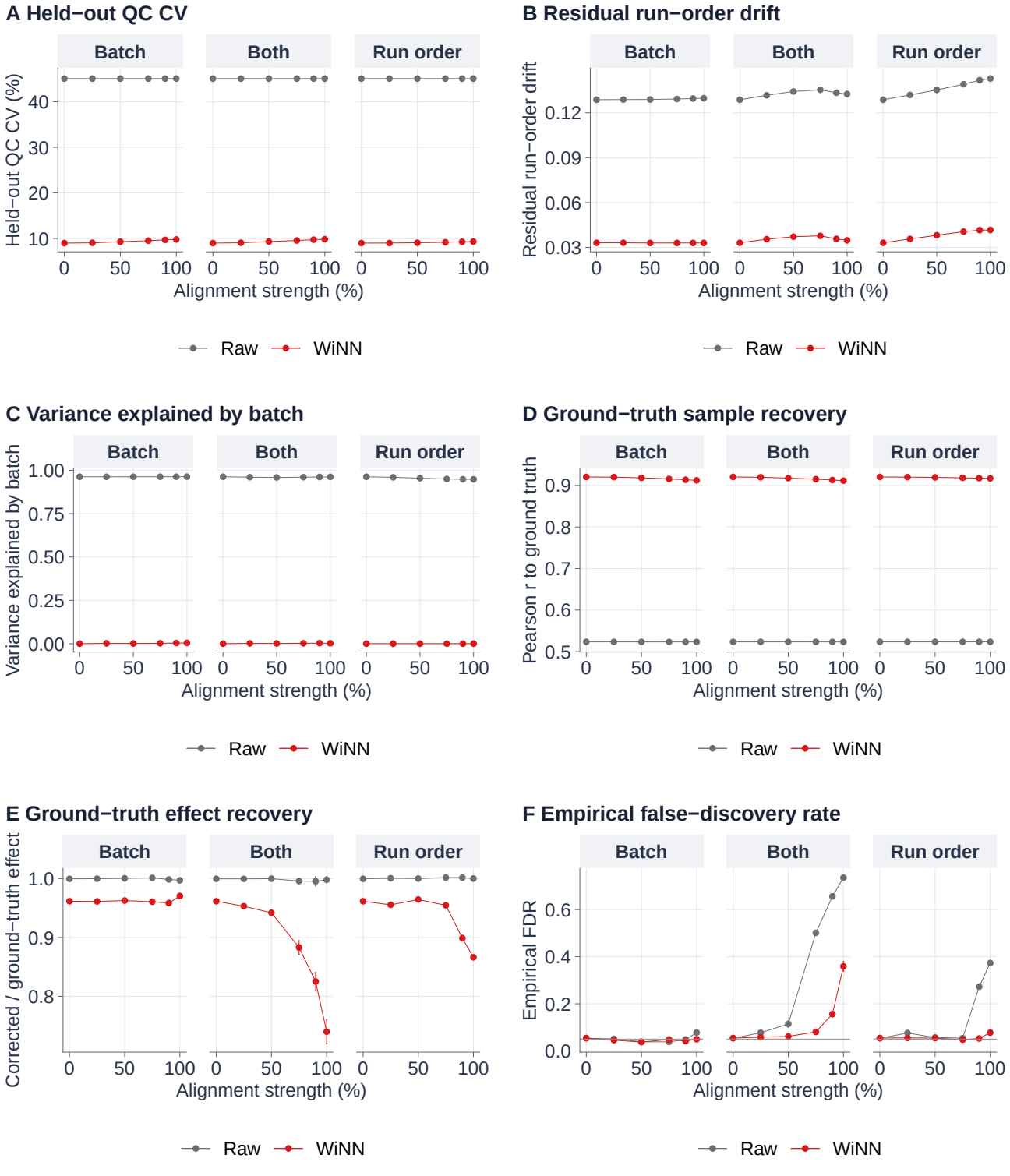

A QC CV improvement
WaveICA
Sacurine
BatchCorr Set 1
Clinical FIA−MS
MTBLS79
Simulation
MAD
Drift
Batch
PQN
−30 −20 −10 0 10 20 30
B Run−order improvement
WaveICA
Sacurine
BatchCorr Set 1
Clinical FIA−MS
MTBLS79
Simulation
MAD
Drift
Batch
PQN
−10 0 10
C Batch−structure improvement
WaveICA
Sacurine
BatchCorr Set 1
Clinical FIA−MS
MTBLS79
Simulation
MAD
Drift
Batch
PQN
−50 0 50


A Held−out QC CV
Simulation
MTBLS79
Clinical
BatchCorr
Sacurine
WaveICA
Held−out QC CV (%)
9.2
9.1
9.0
8.9
17.5
17.0
16.5
16.0
12.6
12.4
12.2
12.0
27.2
27.0
26.8
22
21
20
19
37.7
37.6
37.5
Selective / selective
Forced drift
Forced batch
Forced drift + batch
B Residual run−order drift
Residual run−order drift
0.03
0.02
0.01
0.14
0.12
0.10
0.08
0.06
0.04
0.05
0.04
0.03
0.02
0.01
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0.04
0.03
0.02
0.01
Selective / selective
Forced drift
Forced batch
Forced drift + batch
C Variance explained by batch
Variance explained by batch
9e−04
6e−04
3e−04
0.011
0.010
0.009
0.008
0.00225
0.00200
0.00175
0.00150
0.00125
0.00062
0.00060
0.00058
0.00056
6e−04
4e−04
2e−04
0.0028
0.0026
0.0024
Selective / selective
Forced drift
Forced batch
Forced drift + batch


D Simulation ground−truth RMSE
Selective / selective
Forced drift
Forced batch
Forced drift + batch
0.0725
0.0750
0.0775
Ground−truth RMSE
E Simulation feature ICC(A,1)
Selective / selective
Forced drift
Forced batch
Forced drift + batch
0.915
0.920
ICC(A,1) to ground truth
F Simulation sample correlation
Selective / selective
Forced drift
Forced batch
Forced drift + batch
0.915
0.920
0.925
Pearson r to ground truth

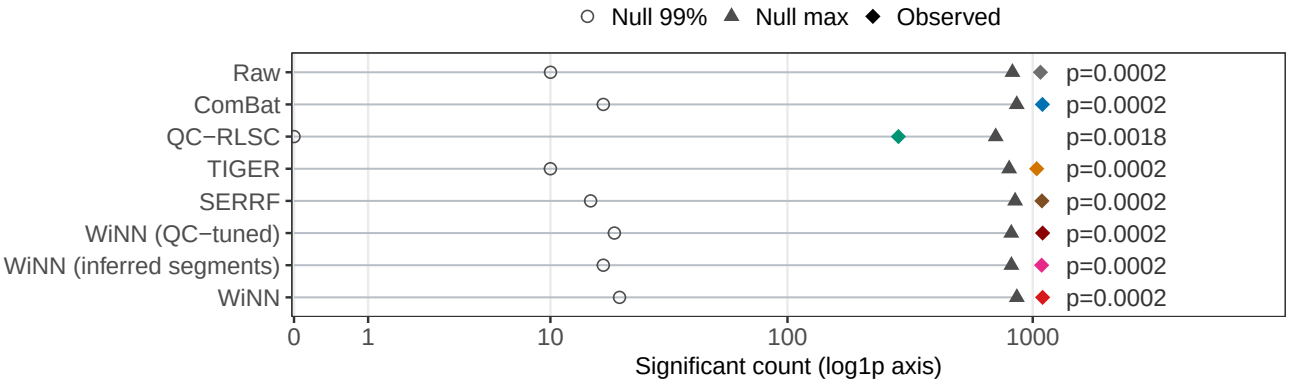

A Held−out QC CV
Raw
ComBat
QC−RLSC
QC−RFSC
TIGER
SERRF
WiNN (QC−tuned)
WiNN (inferred segments)
WiNN
10
20
30
40
Held−out QC CV (%)
B Residual run−order drift
0.05
0.10
0.15
Residual run−order drift
C Variance explained by batch
0.0
0.5
1.0
Variance explained by batch
D Ground−truth RMSE
0.1
0.2
0.3
Ground−truth RMSE
E Feature ICC(A,1)
0.6
0.7
0.8
0.9
ICC(A,1) to ground truth
F Sample correlation
0.5
0.6
0.7
0.8
0.9
Pearson r to ground truth


A Simulation
Raw
ComBat
QC−RLSC
TIGER
SERRF
WiNN (QC−tuned)
WiNN (inferred segments)
WiNN
10
20
30
40
Held−out QC CV (%)
B MTBLS79
0
100
200
300
C Clinical FIA−MS
8
10
12
14
D BatchCorr Set 1
20
30
40
50
60
70
E Sacurine
F WaveICA


A Held−out QC CV
Batch
Both
Run order
Held−out QC CV (%)
40
30
20
10
0
50
100
Alignment strength (%)
Raw
WiNN
B Residual run−order drift
Residual run−order drift
0.12
0.09
0.06
0.03
C Variance explained by batch
Variance explained by batch
1.00
0.75
0.50
0.25
0.00
D Ground−truth sample recovery
Pearson r to ground truth
0.9
0.8
0.7
0.6
0.5
E Ground−truth effect recovery
Corrected / ground−truth effect
1.0
0.9
0.8
F Empirical false−discovery rate
Empirical FDR
0.6
0.4
0.2
0.0

A MTBLS79 cow vs sheep

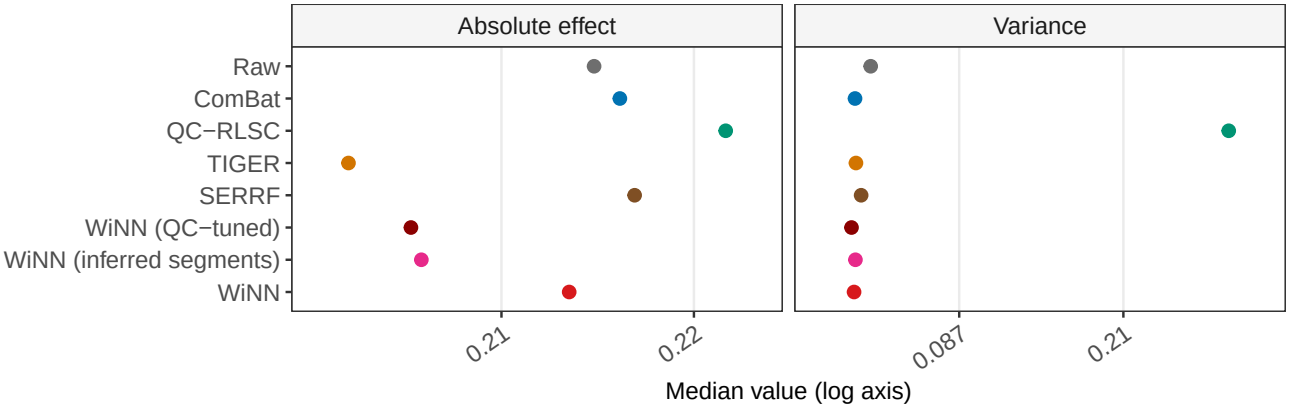
Null tails and observed counts



5,000 constrained permutations.
p = (1 + null counts >= observed) / 5,001.

B MTBLS79 cow vs sheep

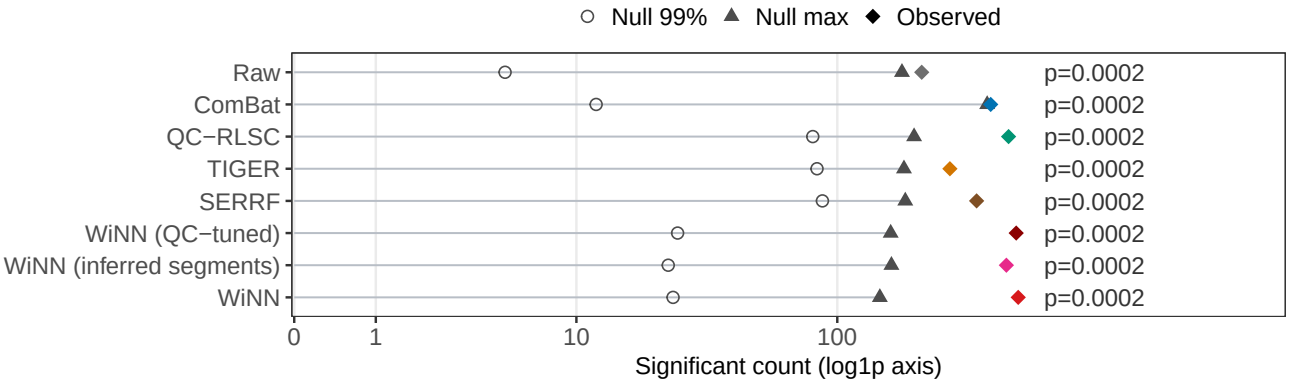
Absolute effects and variances



Effects: log1p intensity scale.

C WaveICA biological groups

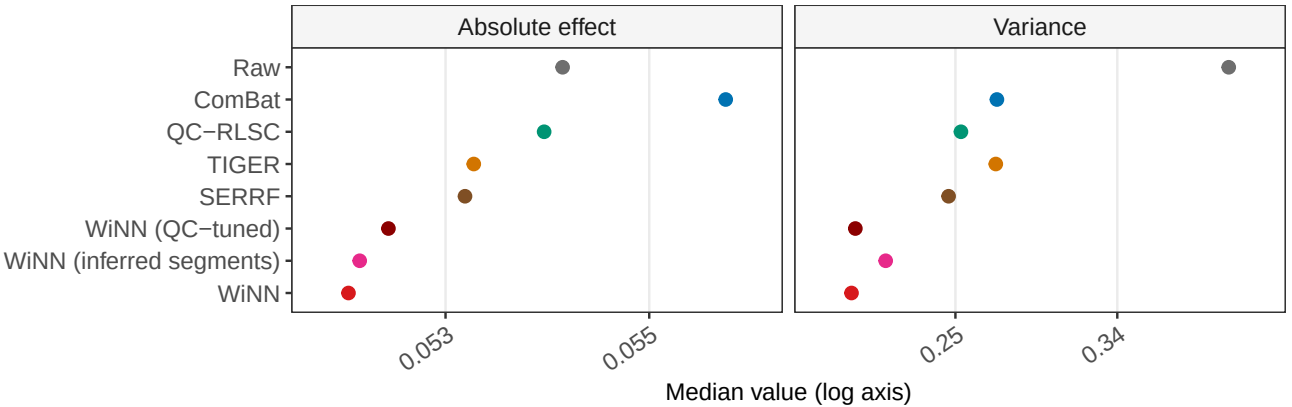
Null tails and observed counts



5,000 constrained permutations.
p = (1 + null counts >= observed) / 5,001.

D WaveICA biological groups

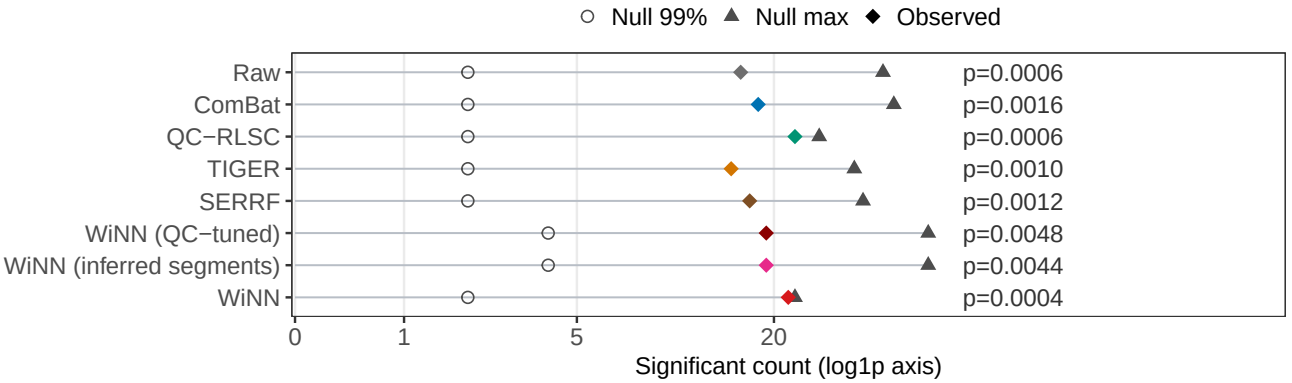
Absolute effects and variances



Effects: log1p intensity scale.

E Sacurine age

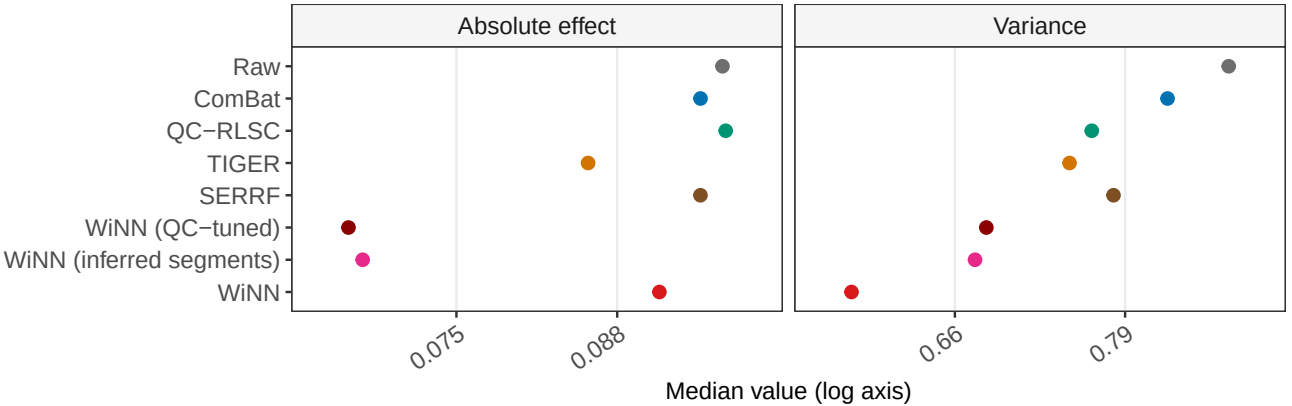
Null tails and observed counts



5,000 constrained permutations.
p = (1 + null counts >= observed) / 5,001.

F Sacurine age

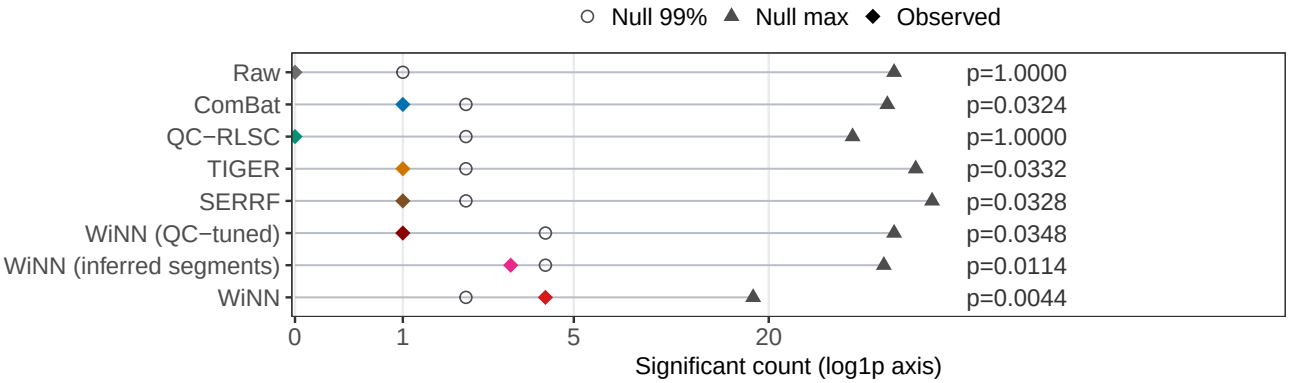
Absolute effects and variances



Effects: log1p after osmolality scaling.

G Sacurine BMI

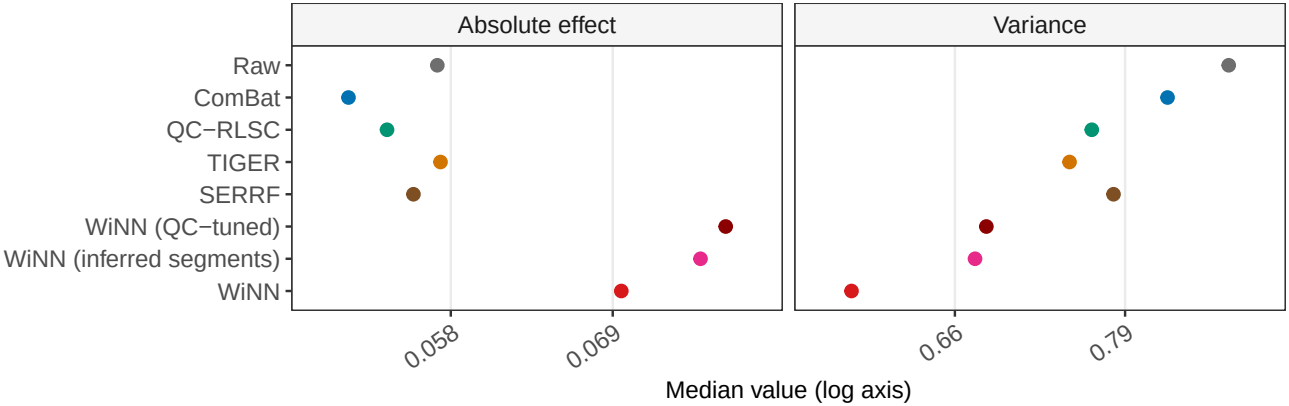
Null tails and observed counts



5,000 constrained permutations.
p = (1 + null counts >= observed) / 5,001.

H Sacurine BMI

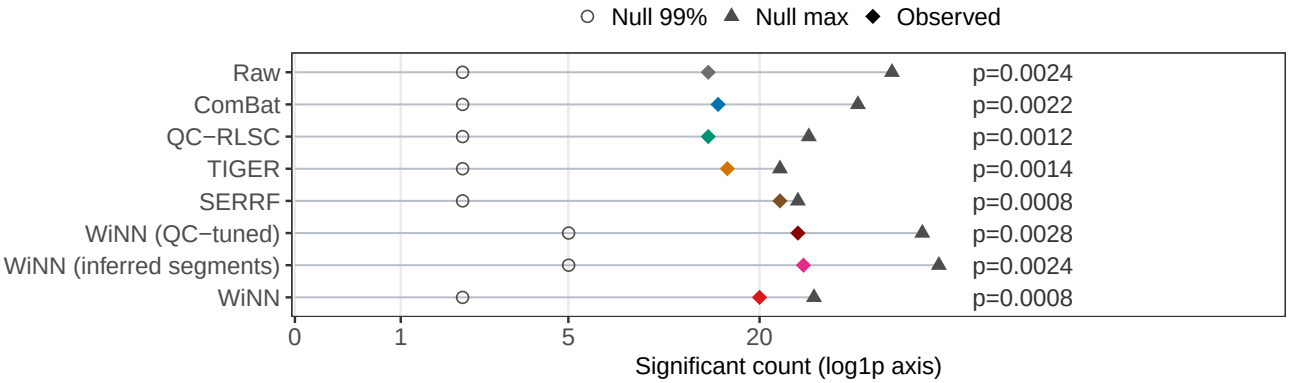
Absolute effects and variances



Effects: log1p after osmolality scaling.

I Sacurine gender

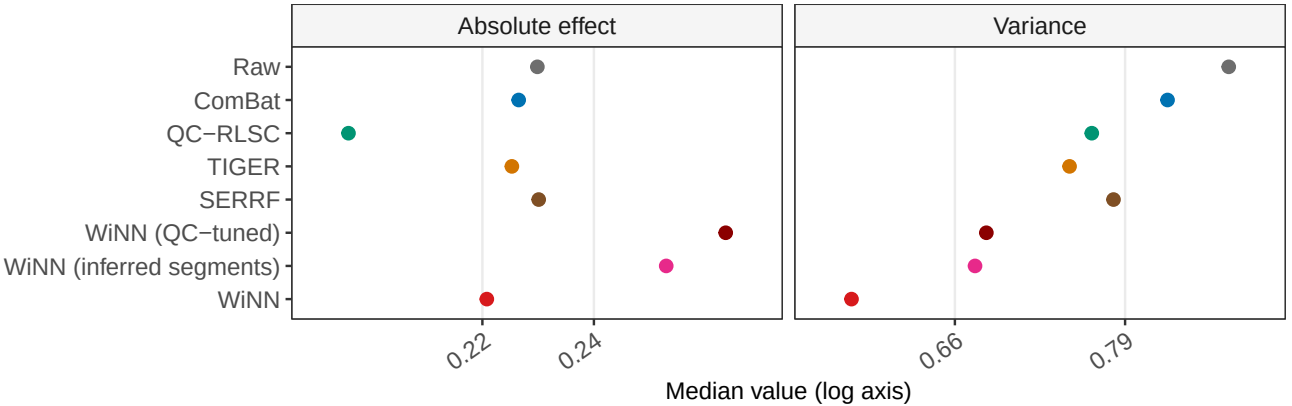
Null tails and observed counts



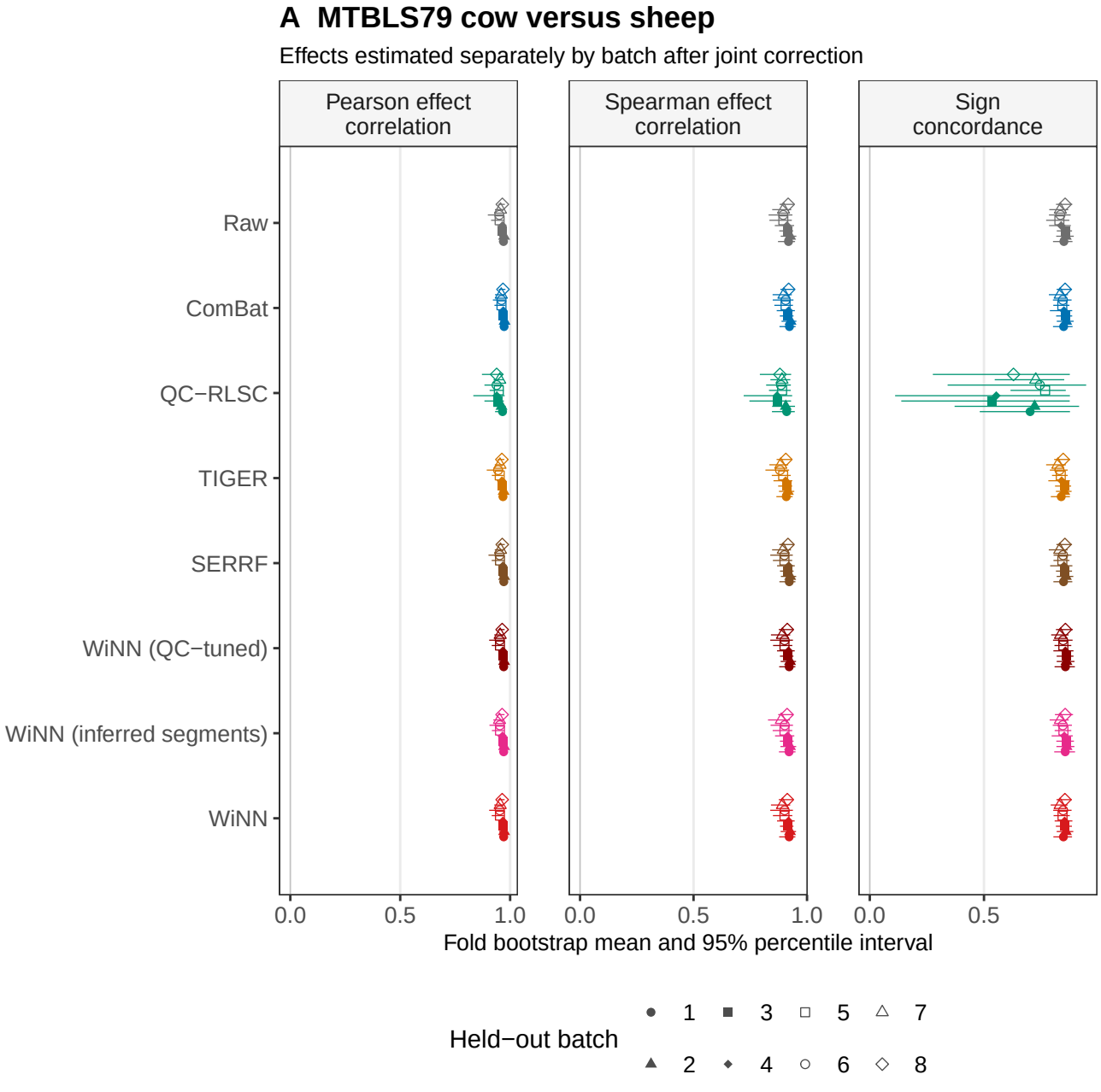
5,000 constrained permutations.
p = (1 + null counts >= observed) / 5,001.

J Sacurine gender

Absolute effects and variances

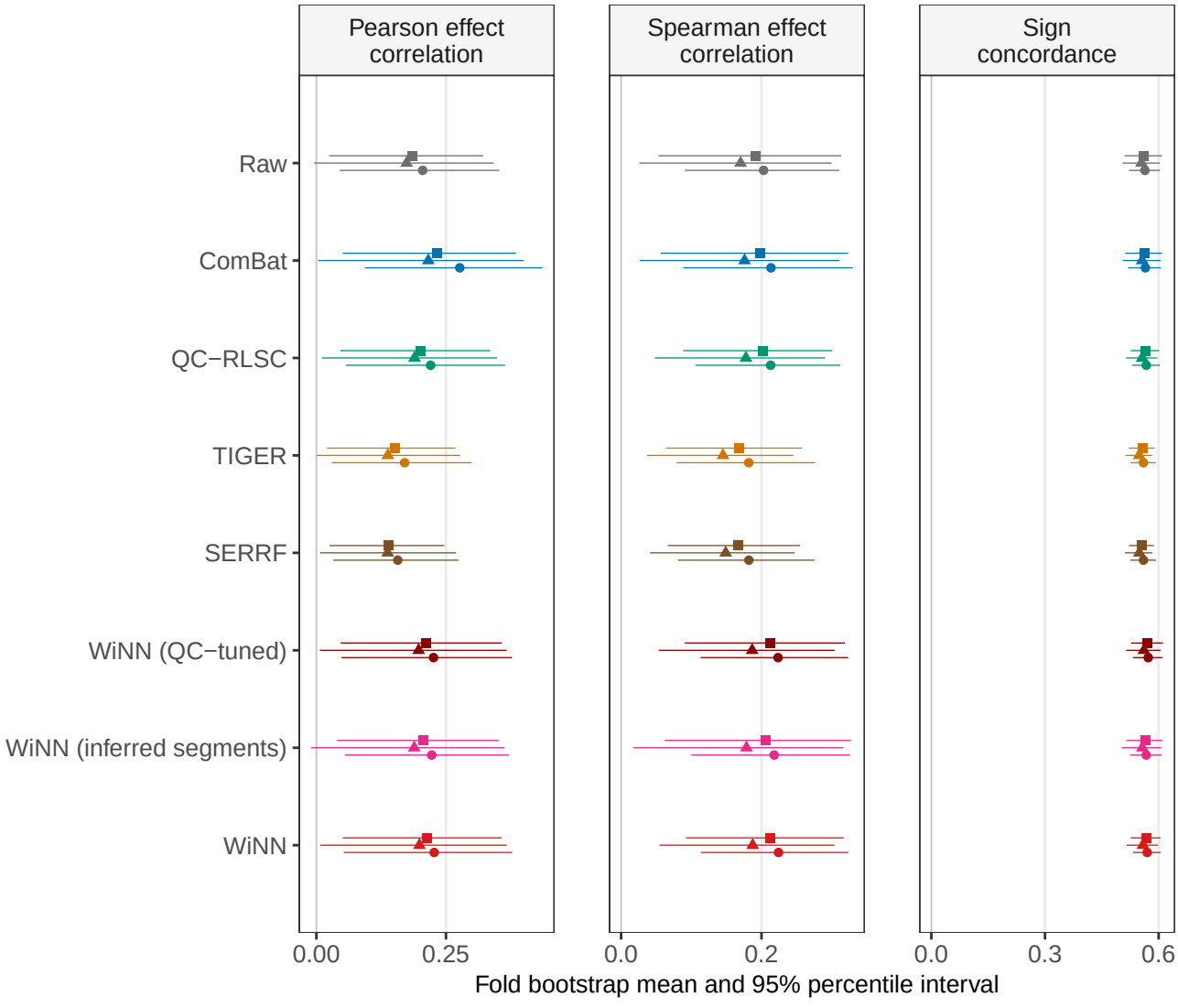


Effects: log1p after osmolality scaling.

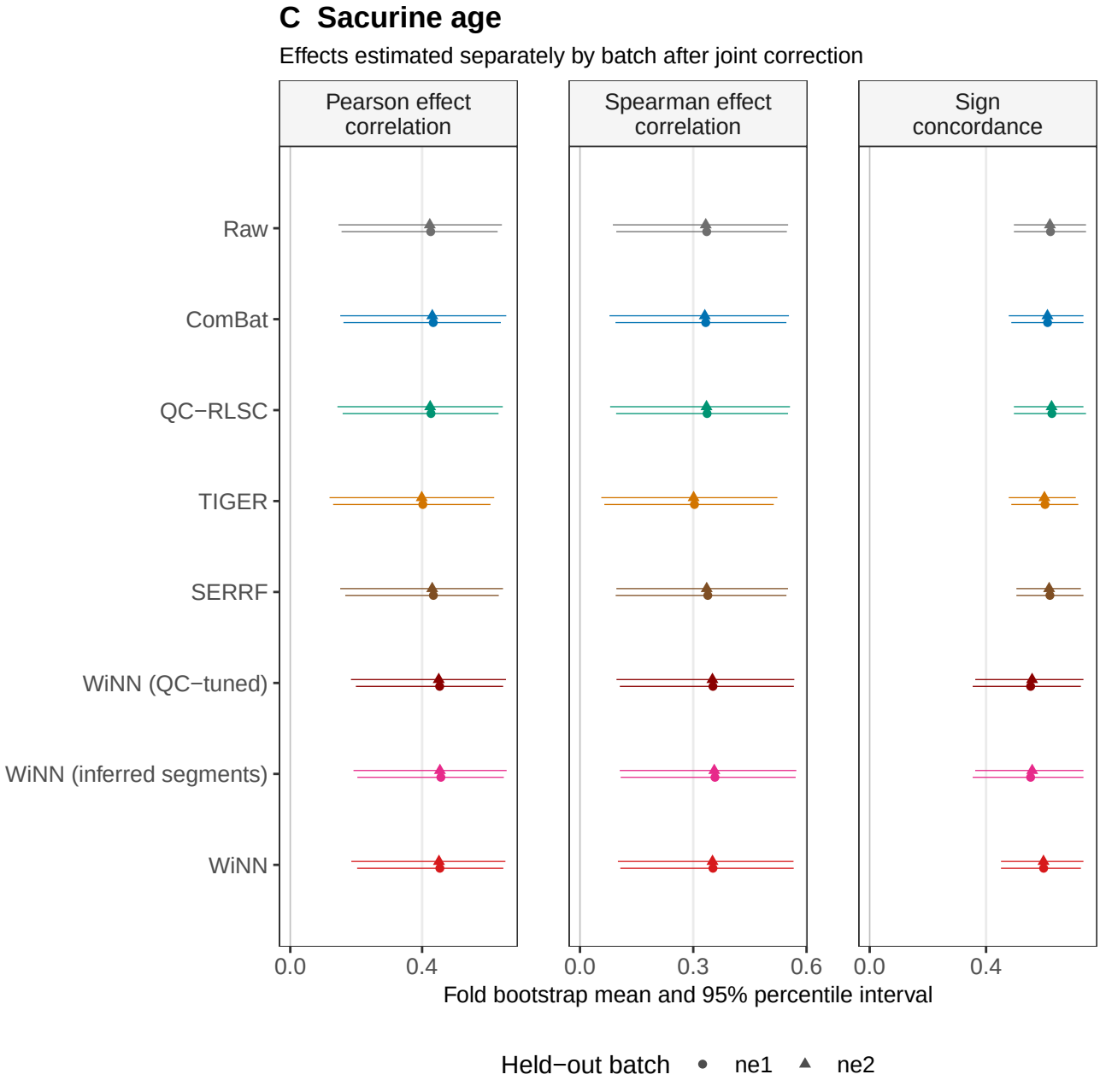

A MTBLS79 cow versus sheep
Effects estimated separately by batch after joint correction
Pearson effect correlation
Spearman effect correlation
Sign concordance
Raw
ComBat
QC−RLSC
TIGER
SERRF
WiNN (QC−tuned)
WiNN (inferred segments)
WiNN
0.0
0.5
1.0
Fold bootstrap mean and 95% percentile interval
Held−out batch
1
2
3
4
5
6
7
8
Intervals describe individual folds after joint correction.

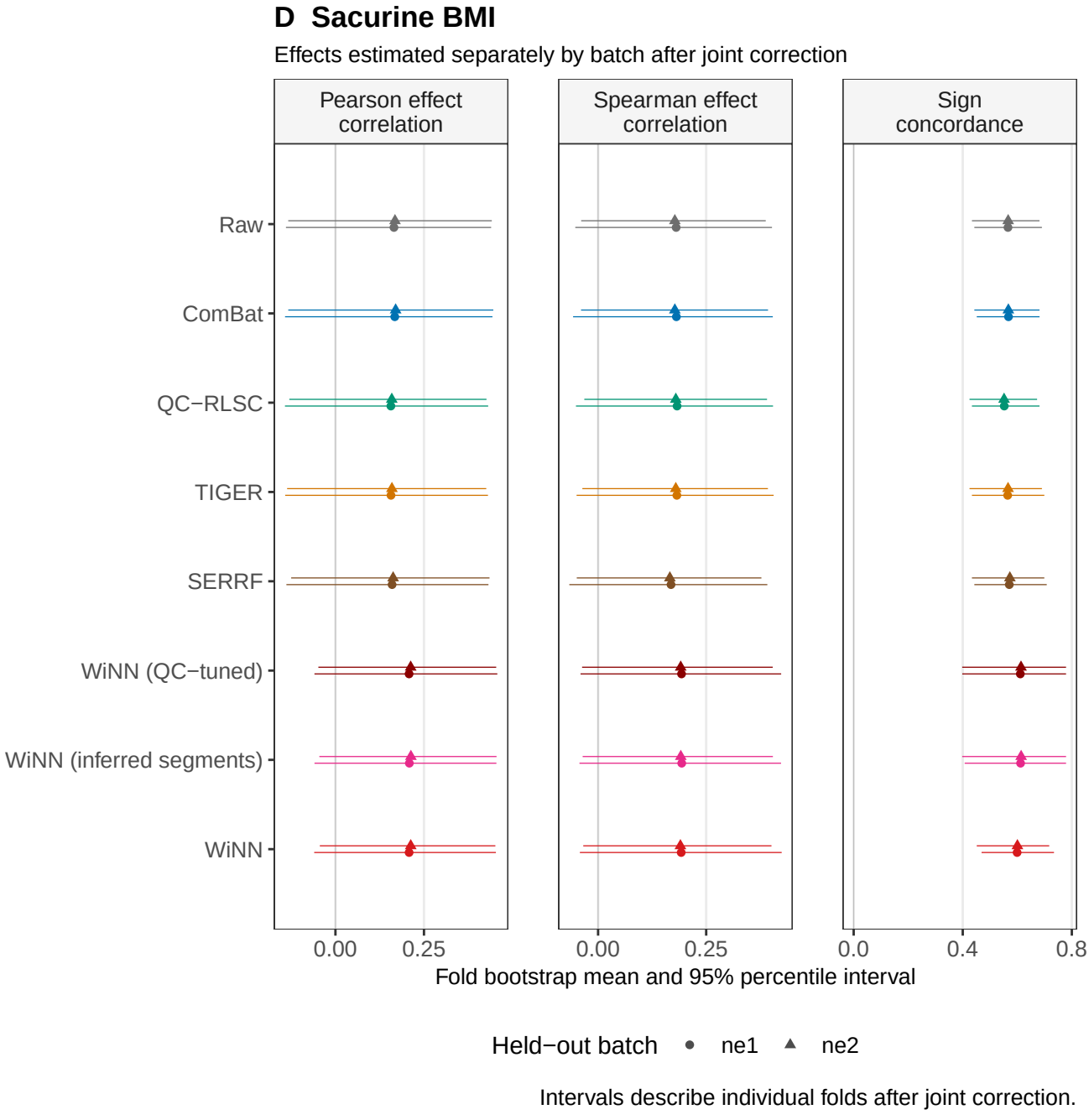
B WaveICA biological groups

Effects estimated separately by batch after joint correction



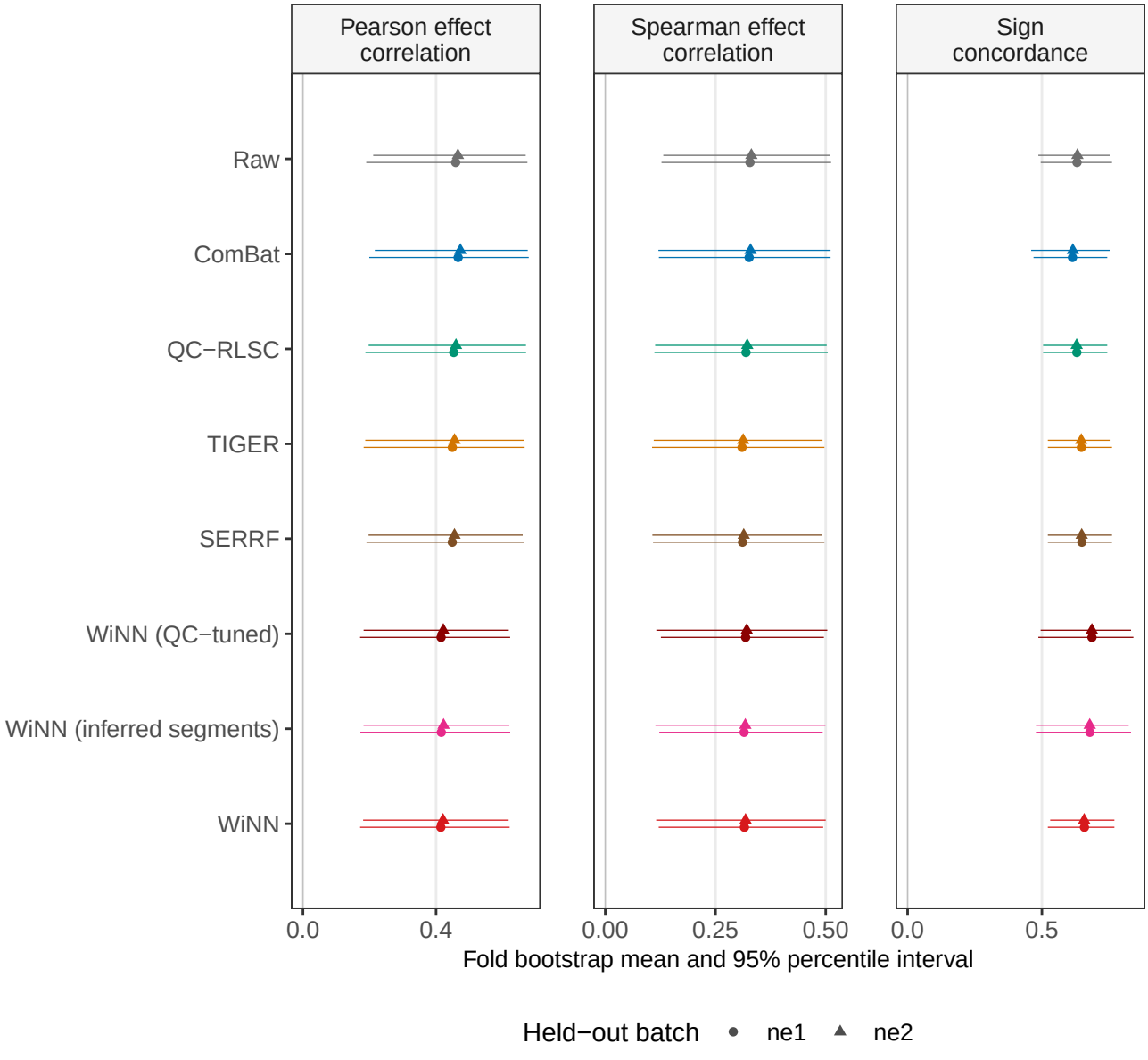
Intervals describe individual folds after joint correction.


C Sacurine age
Effects estimated separately by batch after joint correction
Pearson effect correlation
Spearman effect correlation
Sign concordance
Raw
ComBat
QC−RLSC
TIGER
SERRF
WiNN (QC−tuned)
WiNN (inferred segments)
WiNN
0.0
0.4
0.0
0.3
0.6
0.0
0.4
Fold bootstrap mean and 95% percentile interval
Held−out batch
ne1
ne2
Intervals describe individual folds after joint correction.

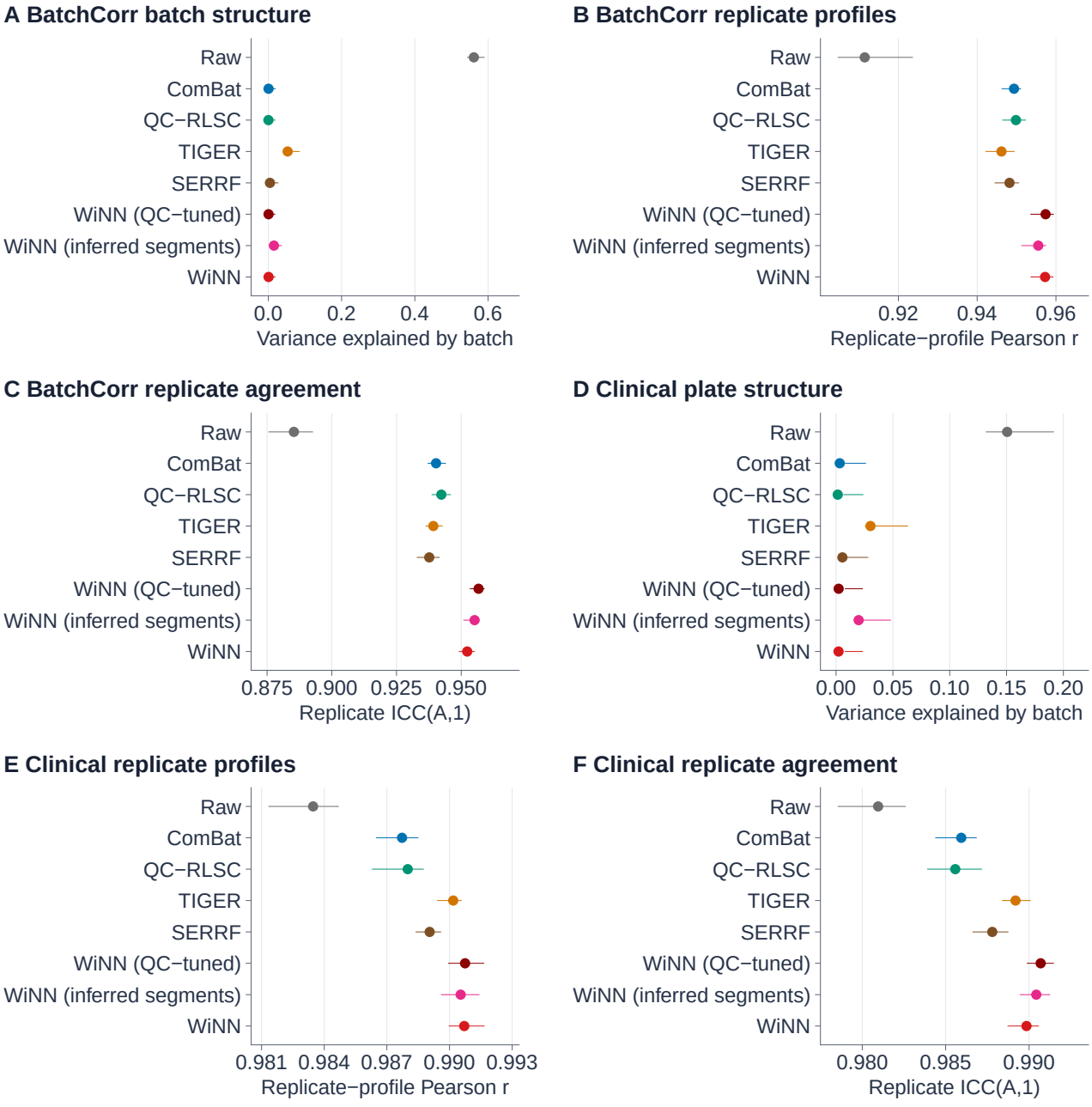

D Sacurine BMI
Effects estimated separately by batch after joint correction
Pearson effect correlation
Spearman effect correlation
Sign concordance
Raw
ComBat
QC−RLSC
TIGER
SERRF
WiNN (QC−tuned)
WiNN (inferred segments)
WiNN
0.00
0.25
0.00
0.25
0.0
0.4
0.8
Fold bootstrap mean and 95% percentile interval
Held−out batch
ne1
ne2
Intervals describe individual folds after joint correction.

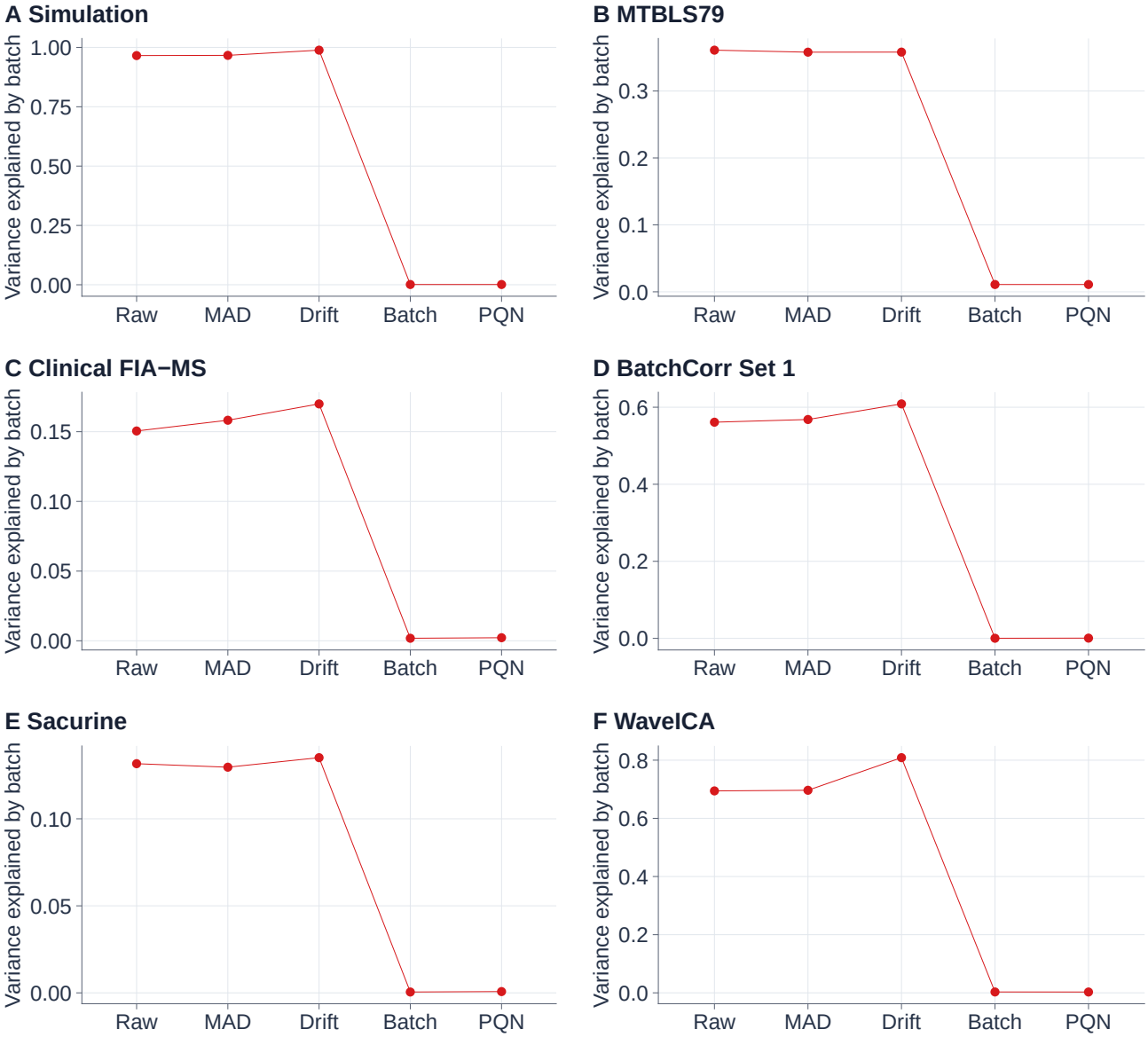
E Sacurine gender

Effects estimated separately by batch after joint correction

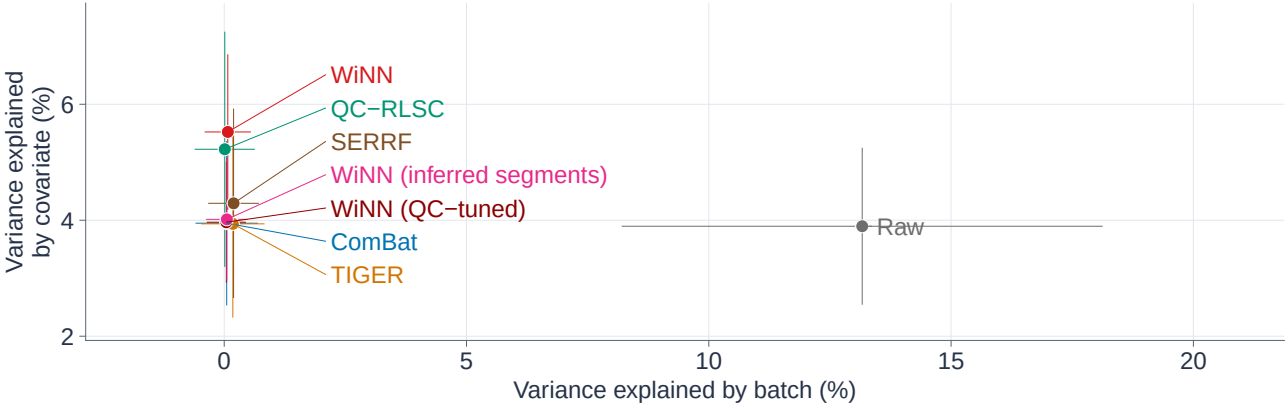


Intervals describe individual folds after joint correction.

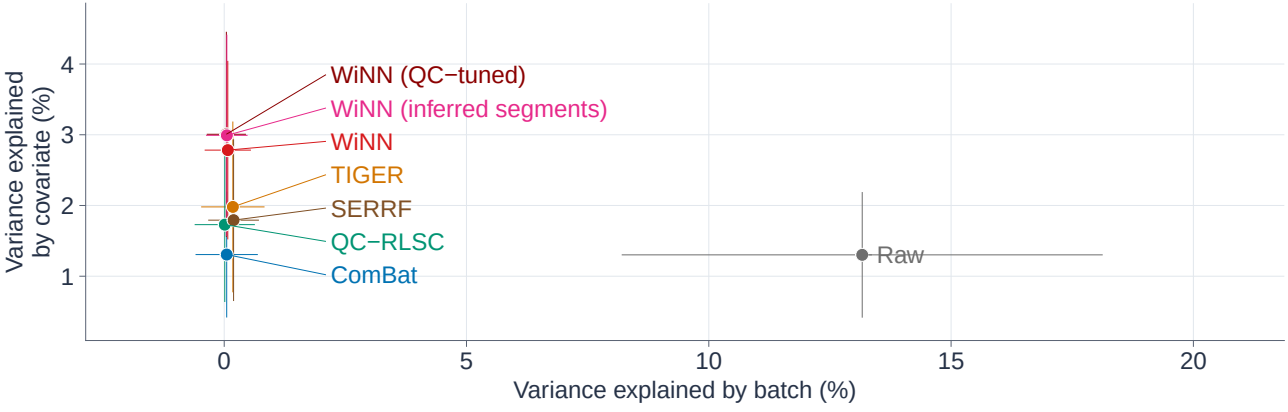

A BatchCorr batch structure
Raw
ComBat
QC−RLSC
TIGER
SERRF
WiNN (QC−tuned)
WiNN (inferred segments)
WiNN
0.0
0.2
0.4
0.6
Variance explained by batch
B BatchCorr replicate profiles
0.92
0.94
0.96
Replicate−profile Pearson r
C BatchCorr replicate agreement
0.875
0.900
0.925
0.950
Replicate ICC(A,1)
D Clinical plate structure
0.00
0.05
0.10
0.15
0.20
Variance explained by batch
E Clinical replicate profiles
0.981
0.984
0.987
0.990
0.993
Replicate−profile Pearson r
F Clinical replicate agreement
0.980
0.985
0.990
Replicate ICC(A,1)


A Simulation
B MTBLS79
C Clinical FIA−MS
D BatchCorr Set 1
E Sacurine
F WaveICA
Variance explained by batch
Raw
MAD
Drift
Batch
PQN

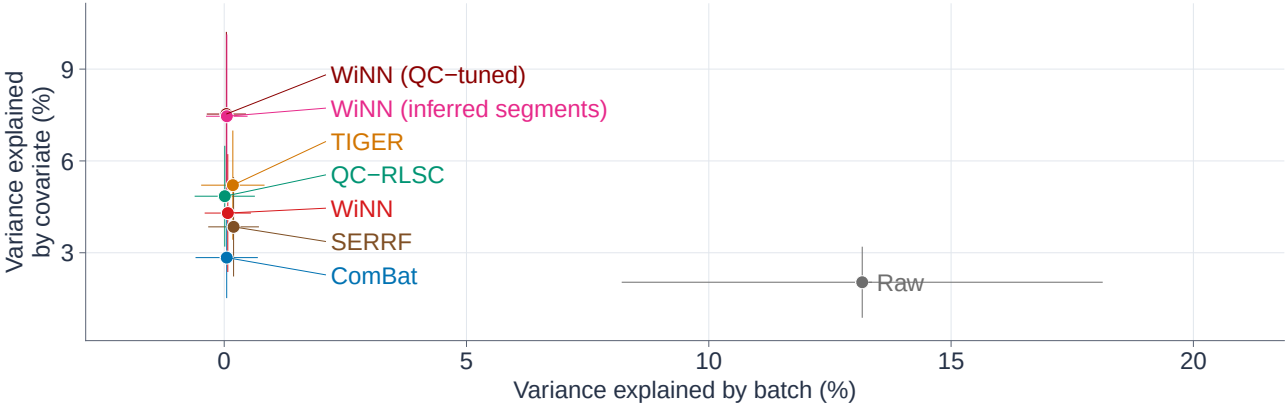
A Sacurine age

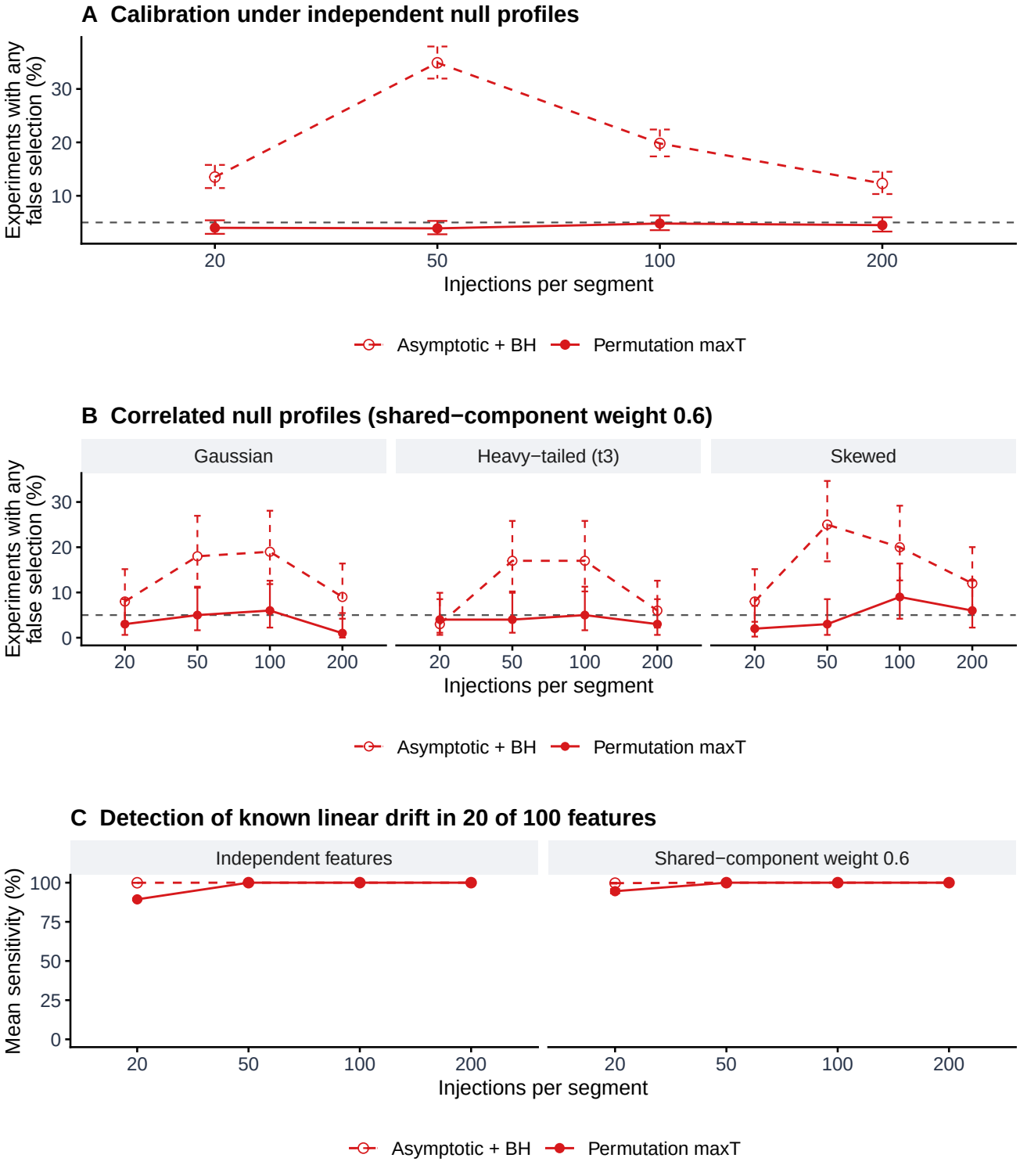


B Sacurine BMI



C Sacurine gender




A Calibration under independent null profiles
Experiments with any false selection (%)
10
20
30
20
50
100
200
Injections per segment
Asymptotic + BH
Permutation maxT
B Correlated null profiles (shared−component weight 0.6)
Gaussian
Heavy−tailed (t3)
Skewed
Experiments with any false selection (%)
0
10
20
30
20
50
100
200
Injections per segment
Asymptotic + BH
Permutation maxT
C Detection of known linear drift in 20 of 100 features
Independent features
Shared−component weight 0.6
Mean sensitivity (%)
0
25
50
75
100
20
50
100
200
Injections per segment
Asymptotic + BH
Permutation maxT